

$$p = 31 \quad n = p \cdot q = 1147$$

$$q = 37$$

$$e = 17$$

$$\phi(n) = (p-1) \cdot (q-1) = 1080$$

$$y = 2$$

$$d: e \cdot d \equiv 1 \pmod{\phi(n)}$$

$\Rightarrow$ Wolfram Alpha (WA)

$$d = 953$$

$$\text{Probe: } 17 \cdot 953 = 16201$$

$$\frac{16201}{1080} = 15,0003... = k$$

$$(k-15) \cdot 1080 = 1 \quad \checkmark$$

$$X = y^d \pmod{n} = 2^{953} \pmod{1147} = 721 //$$

Probe Chiffrieren von x:

$$y = x^e \pmod{n} = 721^{17} \pmod{1147} \stackrel{\text{WA}}{=} 2 //$$